

coal-creek-exploratory

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Packages

```
library(tidyverse)
library(here)
library(janitor)
library(viridis)
library(wesanderson)
```

Data

```
compiled_isotope <- read_csv(here("data", "compiled.csv"), show_col_types =
FALSE) |>
  clean_names() |> # lowercase and underscores in column names
  rename(sample = sample_name, # renaming columns to be shorter
         delta_d = processed_delta_d,
         delta_d_std = processed_delta_d_stddev,
         delta_o = processed_delta_18o,
         delta_o_std = processed_delta_18o_stddev) |>
  mutate(date = mdy(date)) # telling r the date format

gradient_cc <- read_csv(here("data", "cc_gradient_data.csv"))
```

Rows: 5 Columns: 18

— Column

specification —————

Delimiter: ","

chr (13): site, location, Δx (m), cumulative_dams, Δdams, cumulative_dam_len...

dbl (4): total_dams, avg_dam_length, lc-excess_6-23, lc-excess_7-7

num (1): normalized_stream_distance_m

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
coal_creek <- compiled_isotope |> # the sites of focus
  filter(location %in% c("cc")) |>
```

```

filter(site != "coal_3")

coal_creek_all <- compiled_isotope |> # includes coal creek 3 that is a tributary
to main creek
filter(location %in% c("cc"))

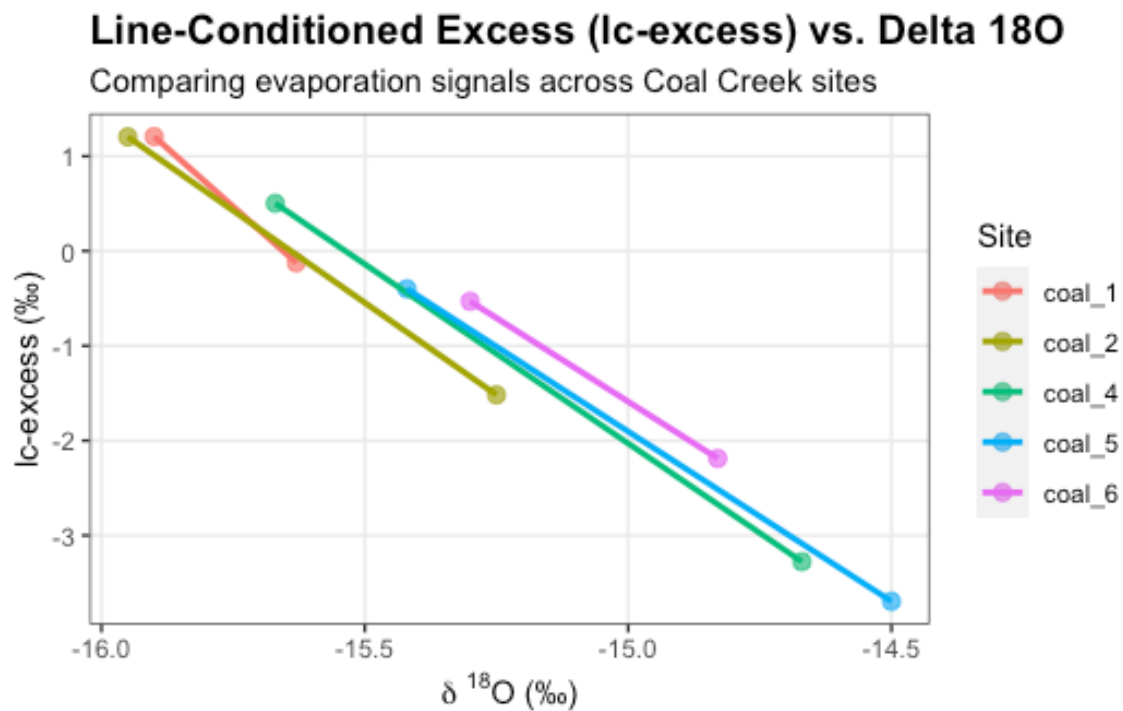
```

Figures

```

ggplot(data = coal_creek, aes(x = delta_o, y = lc_excess, color = site)) +
  geom_point(size = 2.5, alpha = 0.7) +
  geom_smooth(method = "lm", se = TRUE, alpha = 0.15, linewidth = 1) +
  labs(
    title = "Line-Conditioned Excess (lc-excess) vs. Delta 180",
    subtitle = "Comparing evaporation signals across Coal Creek sites",
    x = expression(paste(delta, " ^18, " O (\u2030\u2030))),
    y = "lc-excess (\u2030\u2030)",
    color = "Site") +
  theme_bw() +
  theme(plot.title = element_text(face = "bold", size = 14),
        legend.position = "right",
        panel.grid.minor = element_blank())

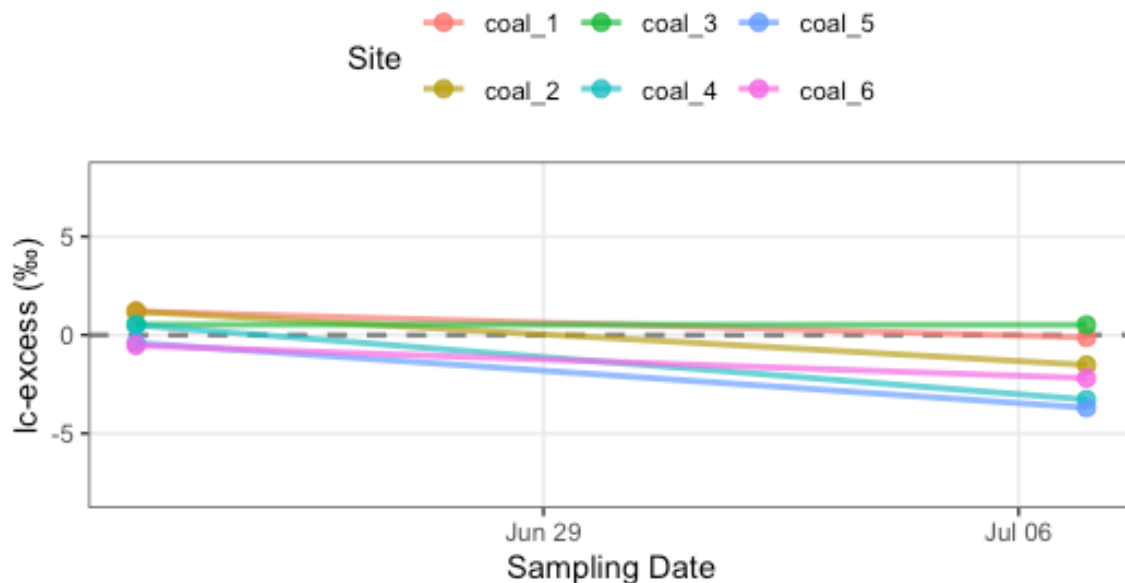
```



```
ggplot(data = coal_creek_all,
       aes(x = date, y = lc_excess, color = site, group = site)) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "gray50", linewidth =
0.8) +
  geom_point(size = 2.5, alpha = 0.8) +
  geom_line(linewidth = 1, alpha = 0.7) +
  labs(title = "Time-Series of Line-Conditioned Excess (lc-excess)",
       subtitle = "Coal Creek -- values below 0 indicate potential evaporative
enrichment",
       x = "Sampling Date",
       y = "lc-excess (‰)",
       color = "Site") +
  ylim(-8, 8) +
  theme_bw() +
  theme(plot.title = element_text(face = "bold", size = 14),
        panel.grid.minor = element_blank(),
        legend.position = "top")
```

Time-Series of Line-Conditioned Excess (lc-excess)

Coal Creek -- values below 0 indicate potential evaporative enrichment

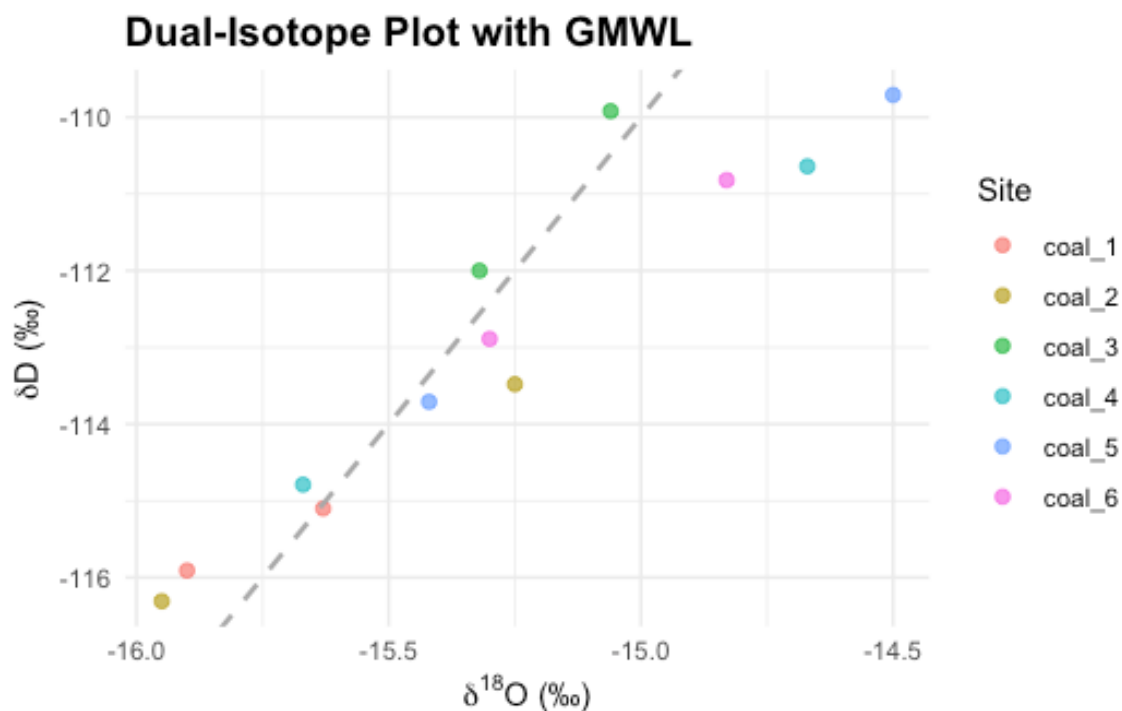


```
ggplot(coal_creek_all, aes(x = delta_o, y = delta_d, color = site)) +
  geom_point(alpha = 0.7, size = 2) +
  # Add Global Meteoric Water Line (GMWL: y = 8x + 10)
  geom_abline(intercept = 10, slope = 8, linetype = "dashed", color = "darkgray",
```

```

linewidth = 0.8) +
  labs(
    title = "Dual-Isotope Plot with GMWL",
    x = expression(paste(delta^{18}, "O (‰)")),
    y = expression(paste(delta, "D (‰)")),
    color = "Site"
  ) +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 14))

```



```

# Coal Creek LMWL calculated from OPIC
lmwl_slope <- 7.93
lmwl_intercept <- 8.97

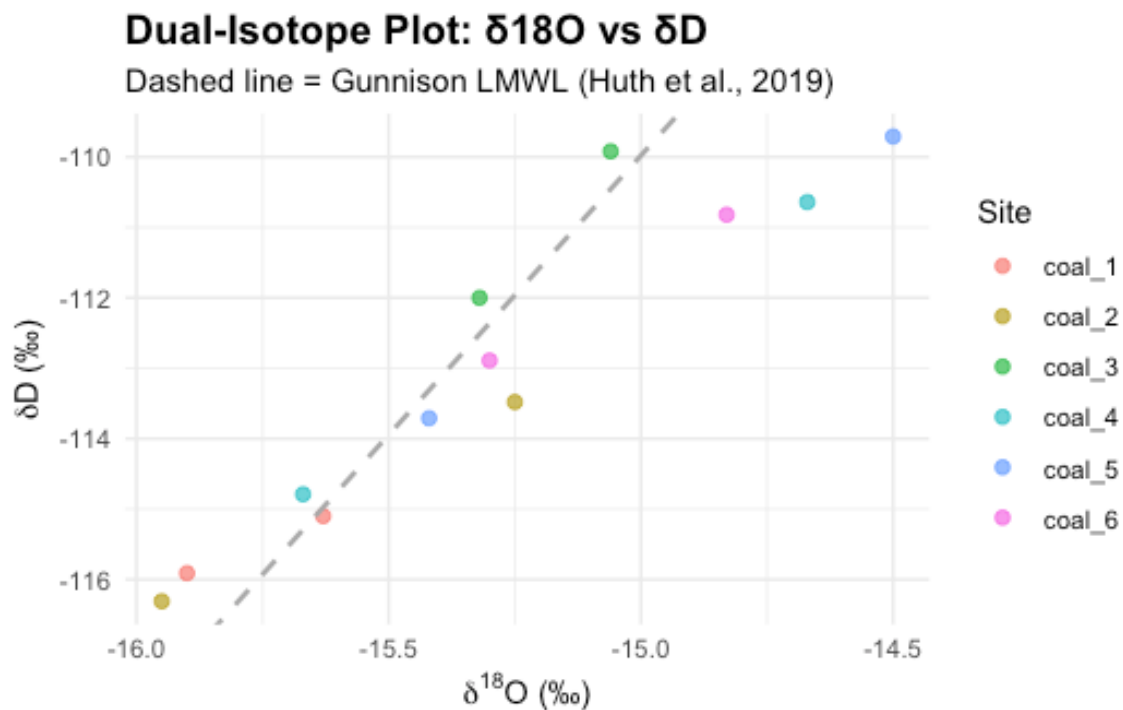
ggplot(coal_creek_all, aes(x = delta_o, y = delta_d, color = site)) +
  geom_point(alpha = 0.7, size = 2) +
  geom_abline(intercept = lmwl_intercept, slope = lmwl_slope,
              linetype = "dashed", color = "darkgray", linewidth = 0.8) +
  labs(
    title = "Dual-Isotope Plot: \u03b4\u00b9\u2078O vs \u03b4\u00b4D",
    subtitle = "Dashed line = Gunnison LMWL (Huth et al., 2019)",
    x = expression(paste(delta^{18}, "O (\u2030)")),

```

```

y = expression(paste(delta, "D (\u2030)")),
color = "Site"
) +
theme_minimal() +
theme(plot.title = element_text(face = "bold", size = 14))

```



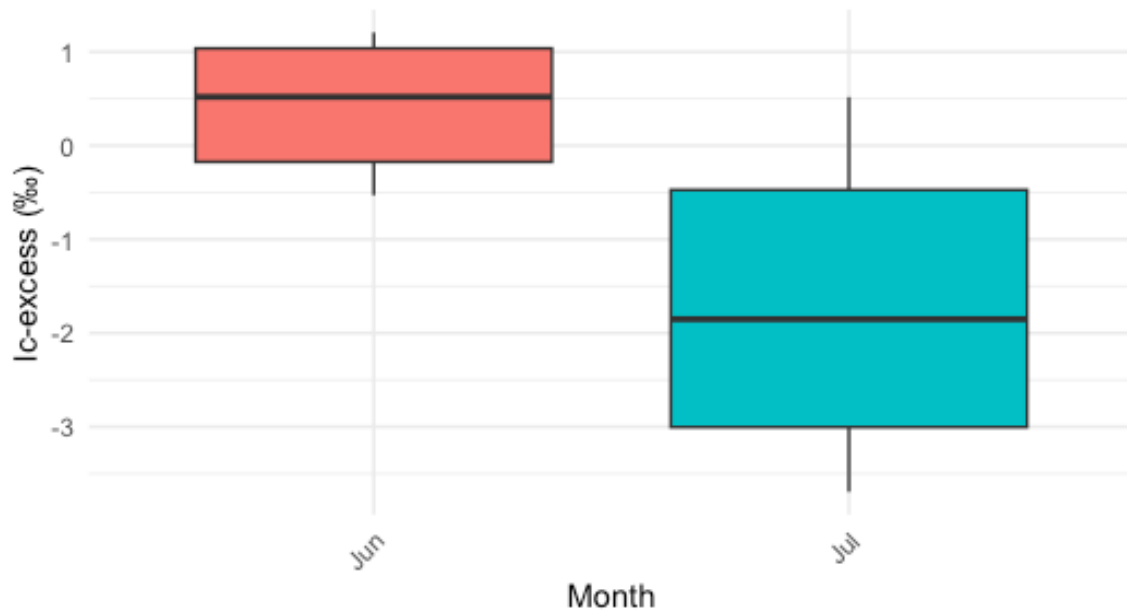
```

ggplot(coal_creek_all, aes(x = factor(month(date, label = TRUE, abbr = TRUE)),
y = lc_excess)) +
geom_boxplot(aes(fill = factor(month(date))), show.legend = FALSE) +
labs(
title = "Monthly Variations in Evaporative Fraction (lc-excess) at Coal Creek",
subtitle = "Seasonal isotopic shifts indicating changes in local evaporation-
condensation histories",
x = "Month",
y = "lc-excess (\u2030)"
) +
theme_minimal() +
theme(plot.title = element_text(face = "bold", size = 14),
axis.text.x = element_text(angle = 45, hjust = 1))

```

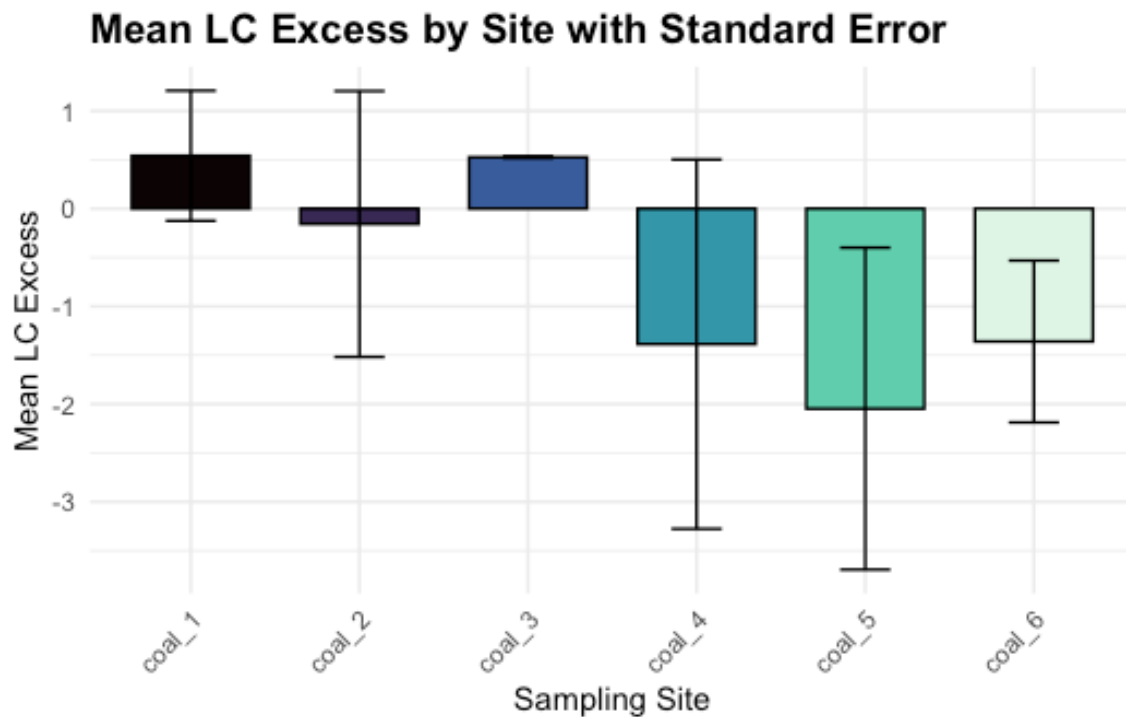
Monthly Variations in Evaporative Fraction (lc-excess)

Seasonal isotopic shifts indicating changes in local evaporation-condensation



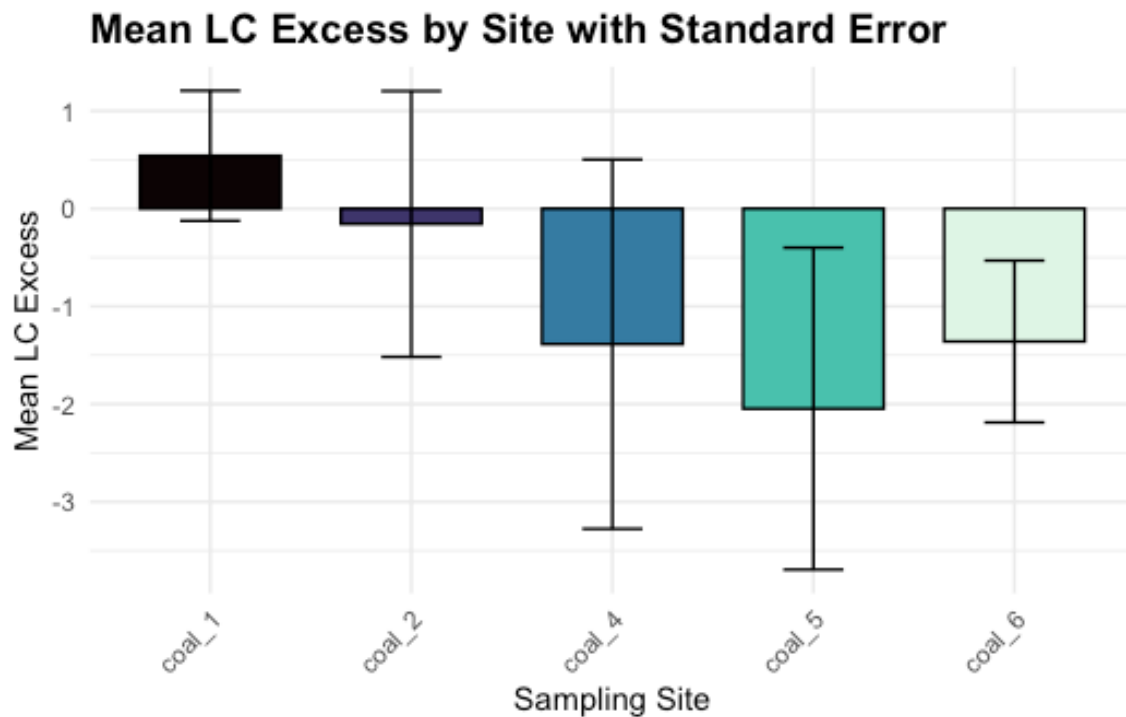
```
delta_summary_cc <- coal_creek_all |>
  group_by(site) |>
  summarise(
    mean_lc = mean(lc_excess, na.rm = TRUE),
    se_lc = sd(lc_excess, na.rm = TRUE) / sqrt(n()),
    .groups = "drop"
  )

ggplot(delta_summary_cc, aes(x = site, y = mean_lc, fill = site)) +
  geom_col(color = "black", width = 0.7) +
  geom_errorbar(aes(ymin = mean_lc - se_lc, ymax = mean_lc + se_lc),
               width = 0.3, color = "black") +
  scale_fill_viridis_d(option = "G") +
  theme_minimal() +
  labs(title = "Mean LC Excess by Site with Standard Error",
       x = "Sampling Site",
       y = "Mean LC Excess") +
  theme(plot.title = element_text(face = "bold", size = 14),
        axis.text.x = element_text(angle = 45, hjust = 1),
        legend.position = "none")
```



```
delta_sum_cc <- coal_creek |>
  group_by(site) |>
  summarise(
    mean_lc = mean(lc_excess, na.rm = TRUE),
    se_lc = sd(lc_excess, na.rm = TRUE) / sqrt(n()),
    .groups = "drop"
  )

ggplot(delta_sum_cc, aes(x = site, y = mean_lc, fill = site)) +
  geom_col(color = "black", width = 0.7) +
  geom_errorbar(aes(ymin = mean_lc - se_lc, ymax = mean_lc + se_lc),
               width = 0.3, color = "black") +
  scale_fill_viridis_d(option = "G") +
  theme_minimal() +
  labs(title = "Mean LC Excess by Site with Standard Error",
       x = "Sampling Site",
       y = "Mean LC Excess") +
  theme(plot.title = element_text(face = "bold", size = 14),
        axis.text.x = element_text(angle = 45, hjust = 1),
        legend.position = "none")
```



```
# Buildings a look up table of stream distances

distances_cc <- tibble(
  site = c("CC1", "CC2", "CC4", "CC5", "CC6"),
  dist_m = c(0, 1496, 2006, 2478, 2885)) |>
  mutate(xmax = dist_m,
         xmin = lag(dist_m, default = 0))

# Creating a filtered data frame for lc-excess
gradient_cc_clean <- gradient_cc |>
  select(site, normalized_stream_distance_m, `Δlc-excess/Δx_6-23`, `Δlc-excess/
Δx_7-7`) |>
  pivot_longer(cols = c(`Δlc-excess/Δx_6-23`, `Δlc-excess/Δx_7-7`),
    names_to = "date",
    values_to = "gradient") |>
  mutate(date = case_when(
    date == "Δlc-excess/Δx_6-23" ~ "2026-06-23",
    date == "Δlc-excess/Δx_7-7" ~ "2026-07-07"),
    gradient = as.numeric(gradient)) |>
  left_join(distances_cc |> select(site, xmin, xmax),
    by = "site") |>
  mutate(site = factor(site, levels = distances_cc$site)) |>
```

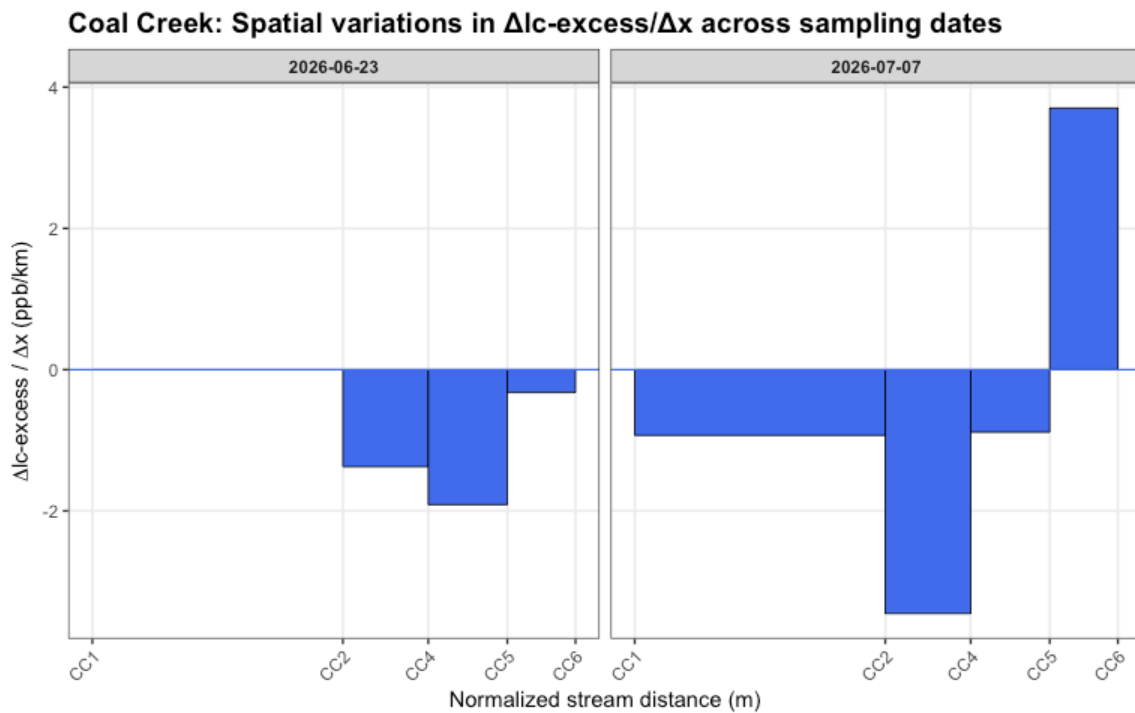


```

filter(!is.na(gradient))

# Gradient plot
ggplot(gradient_cc_clean) +
  geom_rect(aes(xmin = xmin, xmax = xmax, ymin = 0, ymax = gradient),
            fill = "royalblue2", color = "black", linewidth = 0.3) +
  geom_hline(yintercept = 0, color = "royalblue2", linewidth = 0.4) +
  facet_wrap(~date, nrow = 1) +
  scale_x_continuous(
    breaks = distances_cc$dist_m,
    labels = distances_cc$site) +
  labs(
    title = "Coal Creek: Spatial variations in  $\Delta\text{lc-excess}/\Delta x$  across sampling
dates",
    x = "Normalized stream distance (m)",
    y = expression(paste(Delta, "lc-excess / ", Delta, "x (ppb/km)"))) +
  theme_bw(base_size = 11) +
  theme(
    plot.title = element_text(face = "bold", size = 14),
    panel.grid.minor = element_blank(),
    strip.text = element_text(face = "bold"),
    axis.text.x = element_text(angle = 45, hjust = 1))

```



```

gradient_cum_cc <- gradient_cc |>
  select(site, cumulative_dams, `Δlc-excess/Δavg_dam_length_6-23 (ppb/m)`,
    `Δlc-excess/Δcum_dam_length_7-7 (ppb/m)` ) |>
  pivot_longer(
    cols = c(`Δlc-excess/Δavg_dam_length_6-23 (ppb/m)`, `Δlc-excess/
Δcum_dam_length_7-7 (ppb/m)`),
    names_to = "date", values_to = "gradient") |>
  mutate(date = case_when(date == "Δlc-excess/Δavg_dam_length_6-23 (ppb/m)" ~
"2026-06-23",
                        date == "Δlc-excess/Δcum_dam_length_7-7 (ppb/m)"
~ "2026-07-07"),
    gradient = as.numeric(gradient), cumulative_dams =
as.numeric(cumulative_dams),
    site = factor(site)) |>
  filter(!is.na(gradient))

ggplot(gradient_cum_cc, aes(x = site, y = gradient, fill = site)) +
  geom_col(color = "black", width = 0.7) +
  geom_hline(yintercept = 0, color = "black", linewidth = 0.4) +
  facet_wrap(~ date, nrow = 1) +
  scale_fill_manual(values = wes_palette("FantasticFox1", n = 4, type =
"continuous")) +
  scale_x_discrete(labels = function(x) {
    lookup <- setNames(gradient_cum_cc$cumulative_dams, gradient_cum_cc$site)
    lookup[x]
  }) +
  labs(
    x = "Cumulative dam count",
    y = expression(Delta * "lc-excess/" * Delta * "cumulative dam length (" *
"\u2030" * ")/m")
  ) +
  theme_bw(base_size = 12) +
  theme(
    strip.text = element_text(face = "bold", size = 11),
    strip.background = element_rect(fill = "gray85", color = NA),
    panel.grid.minor = element_blank(),
    panel.grid.major.x = element_blank(),
    legend.title = element_text(face = "bold")
  )

```

